MS&E 228 (Winter 2026) — Lecture 16 Notes

Proximal Causal Inference

Vasilis Syrgkanis
Stanford University

March 8, 2026

Readings. *Applied Causal Inference Powered by ML and AI*, Chapter 14. Key papers: Miao, Geng, & Tchetgen Tchetgen (2018), Tchetgen Tchetgen et al. (2020), Cui, Pu, Shi, Miao, & Tchetgen Tchetgen (2024).

1 Introduction: Beyond Bounds

This chapter introduces a powerful and relatively recent identification strategy for dealing with unobserved confounding: **proximal causal inference**. In the previous two chapters, we developed a progressively richer toolkit for settings where conditional exogeneity fails. First, sensitivity analysis allowed us to produce *bounds* on the true treatment effect by making explicit assumptions about the strength of unobserved confounding. Second, instrumental variables (IV) provided a way to *point-identify* a causal effect—the Local Average Treatment Effect (LATE) for compliers—by exploiting an exogenous source of variation in the treatment. The front-door criterion offered yet another route, leveraging a complete, unconfounded mediator.

Proximal causal inference represents a distinct and complementary approach. Rather than requiring an exogenous instrument or a complete mediator, it leverages **incomplete proxies** of the unobserved confounder. The key insight, which has generated significant interest in the empirical literature over the past five to six years, is that if one can find two distinct types of incomplete proxies—a *treatment proxy* and an *outcome proxy*—then the causal effect can be point-identified. This is a remarkable result: variables that individually would only allow for sensitivity bounds can, when combined, yield a precise causal estimate.

The goals of this chapter are:

1. **Motivate proximal inference** by distinguishing complete from incomplete proxies and explaining why two types of incomplete proxies are needed.
2. **Derive the identification result** in the linear setting, showing how the causal effect is recovered through a substitution trick and a system of moment equations.
3. **Connect proximal inference to instrumental variables,** revealing that the proximal estimator is a specific instance of a generalized IV regression.

4. **Develop a practical DML-Proximal estimator** with cross-fitting, influence-function-based inference, and support for overidentification.

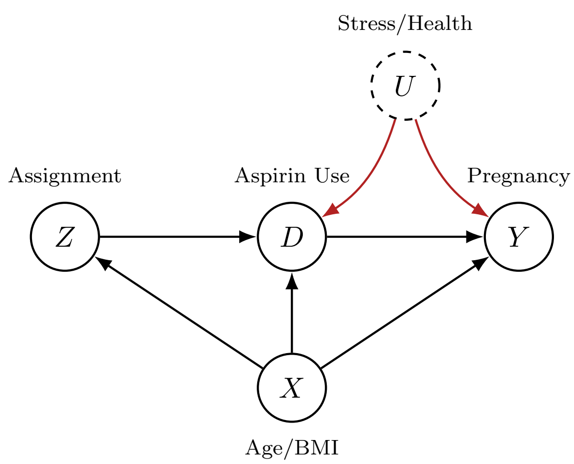
5. **Sketch the non-parametric extension,** introducing the Non-Parametric IV (NPIV) problem and modern solution methods such as DeepIV and adversarial approaches.

2 Recap: The Instrumental Variable Approach

Before introducing proximal causal inference, it is helpful to briefly revisit the instrumental variable framework from the previous chapter, as the two methods share deep structural connections that will become apparent later.

2.1 The IV Setup and Key Conditions

Recall that an instrumental variable Z is a variable that affects the treatment D but has no direct effect on the outcome Y other than through its influence on D. Formally, the IV approach requires three conditions: (i) **relevance**—the instrument Z must have a non-trivial effect on the treatment D; (ii) **exclusion**—Z must not affect Y directly, so that the interventional outcome when we intervene on both D and Z, i.e. $Y(d,z)$, is equal to the interventional outcome when we only intervene on D, i.e. $Y(d,z) = Y(d)$; and (iii) **conditional exogeneity**—the instrument must be as-if randomly assigned, conditional on observable X, meaning there is no unobserved confounding between Z and Y.



Under these conditions, the LATE theorem establishes that in the case of a binary instrument and a binary treatment, if we further assume **monotonicity**, i.e., that the instrument can only encourage but not discourage treatment ($D(Z=1) \geqslant D(Z=0)$), then the ratio of the intent-to-treat effect on the outcome to the intent-to-treat effect on the treatment (compliance) identifies the average treatment effect among compliers:

$$\text{LATE} = \frac{\text{ATE}(Z \to Y)}{\text{ATE}(Z \to D)}$$

Under conditional exogeneity both the numerator and the denominator can be identified by the g-formula:

$$\text{LATE} = \frac{\mathbb{E}[\mathbb{E}[Y \mid Z = 1, X] - \mathbb{E}[Y \mid Z = 0, X]]}{\mathbb{E}[\mathbb{E}[D \mid Z = 1, X] - \mathbb{E}[D \mid Z = 0, X]]}$$

Under the stronger assumption that the outcome is **partially linear** in the treatment—that is, $Y = \theta D + g(X) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid D, X] \neq 0$ but $\mathbb{E}[\varepsilon \mid Z, X] = 0$—the IV approach identifies the Average Treatment Effect θ. The estimation proceeds by first residualizing all variables with respect to the observed controls X, and then solving the moment condition:

$$\mathbb{E}[(\tilde{Y} - \theta\tilde{D}) \cdot \tilde{Z}] = 0,$$

where $\tilde{Y}$, $\tilde{D}$, and $\tilde{Z}$ denote the residualized versions of Y, D, and Z, respectively. The key feature of this moment condition is that the residual from the linear model of $\tilde{Y}$ on $\tilde{D}$ must be uncorrelated with the *instrument* $\tilde{Z}$, rather than with the regressor $\tilde{D}$ itself. This should be contrasted with the OLS estimator, which as we have seen from its normal equations, is trying to find a linear model $\theta\tilde{D}$, such that the residuals are uncorrelated with $\tilde{D}$, i.e., solve the equation $\mathbb{E}[(\tilde{Y} - \theta\tilde{D})\tilde{D}] = 0$. This distinction from ordinary least squares will reappear in a generalized form in the proximal setting.

2.2 Practice: Identifying the Instrument

To build fluency with the IV framework, consider two examples that illustrate how to identify instruments in practice.

Example 1: The Oregon Medicaid Lottery

The state of Oregon expanded Medicaid eligibility via a *lottery*: eligible residents entered a random drawing, and winners could apply for coverage. The goal is to estimate the causal effect of having Medicaid (D) on emergency room visits (Y). The unobserved confounder U represents socioeconomic and health factors that influence both the decision to obtain Medicaid and the frequency of ER visits.

In this setting, the instrument Z is the lottery outcome—whether an individual won the lottery or not. The lottery is randomly assigned (exogenous), it shifts the probability of obtaining Medicaid (relevant), and it has no plausible direct effect on ER visits except through its effect on coverage (exclusion). The **compliers** are those individuals who obtain Medicaid if and only if they win the lottery. The LATE identifies the causal effect for this subgroup.

Example 2: A Travel Website Encouragement A/B Test

A travel website wants to estimate the causal effect of users becoming premium members (D) on downstream engagement (Y). Users who are more engaged are also more likely to become members, creating confounding through unobserved engagement levels (U). The company runs an A/B test: a random subset of users is shown an easier sign-up flow.

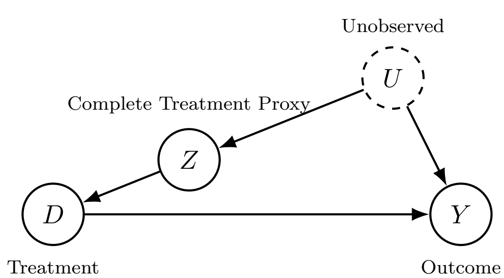
The instrument Z is the assignment to the easier sign-up flow. It is randomly assigned (exogenous), it increases the probability of becoming a member (relevant), and—provided the sign-up prompt is unobtrusive—it should not directly affect engagement except through membership (exclusion). A natural question is whether the exclusion restriction is plausible: if the sign-up flow is very intrusive and appears on every page, it could directly affect user engagement, by irritating the user, violating the exclusion restriction. The compliers are users who become members only when exposed to the easier flow.

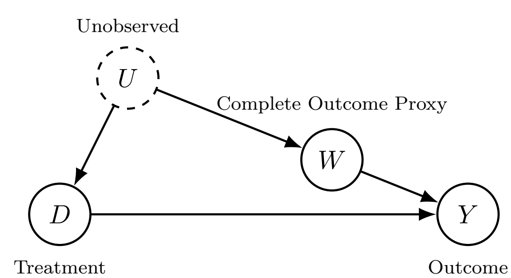
3 From Complete to Incomplete Proxies

With the IV framework fresh in mind, we now turn to a fundamentally different approach. The motivating question is: what can we do when we have variables that are correlated with the unobserved confounder, but that do not satisfy the stringent requirements of a complete mediator of the unobserved confounder on either the treatmet or the outcome (complete proxies)?

3.1 Complete Proxies: When Controlling Suffices

Consider the general setting where a treatment D affects an outcome Y, but there is an unobserved confounder U that influences both. A natural idea is to find a variable Z in the data that is correlated with U and to control for it. If Z is a **complete proxy**—meaning it fully mediates the influence of U on either the treatment or the outcome—then controlling for Z is sufficient to block the confounding path.





In the left graph, Z completely mediates the effect of U on D: the confounder U influences the treatment *only* through Z. In the right graph, W completely mediates the effect of U on Y. In either case, controlling for the complete proxy blocks the confounding path and restores conditional exogeneity.

As a concrete example, consider estimating the effect of a prenatal vitamin (D) on birth outcomes (Y), where the unobserved confounder U represents genetic risk factors. If a patient's decision to take the vitamin is based *solely* on their family's history of birth defects (Z), and not on any direct knowledge of their own genetic factors, then family history is a complete proxy for the genetic factors' influence on the treatment decision. Controlling for family history would suffice to establish conditional exogeneity.

3.2 Incomplete Proxies: The Realistic Case

In practice, however, proxies are almost always **incomplete**. The unobserved confounder retains a direct influence on the treatment or outcome that is not captured by the proxy. In the graphs below, the red arrows indicate the direct paths from U to D (or from U to Y) that bypass the proxy:

Returning to our prenatal vitamin example, family history becomes an incomplete proxy if the physician's own clinical assessment of the patient's genetic risk—information that is part of U but not captured by family history alone—also influences the treatment decision. In this case, controlling for family history reduces confounding bias but does not eliminate it.

When proxies are incomplete, one option is to use **sensitivity analysis**. By controlling for a strong but incomplete proxy, the residual confounding that U can explain is reduced, leading to tighter sensitivity bounds. However, this still yields bounds rather than a point estimate, and the analyst must still make assumptions about the sensitivity parameters.

3.3 The Central Question

This leads to the central question of this chapter:

*Can we leverage incomplete proxies to obtain a **point estimate**—not just bounds—of the causal effect?*

The answer, remarkably, is **yes**—provided we can find *two types* of incomplete proxies simultaneously. This is the core insight of proximal causal inference.

4 The Key Insight: Two Types of Proxies

The fundamental idea behind proximal causal inference is that point identification of the causal effect is possible when two distinct types of incomplete proxies are available simultaneously. These two types are:

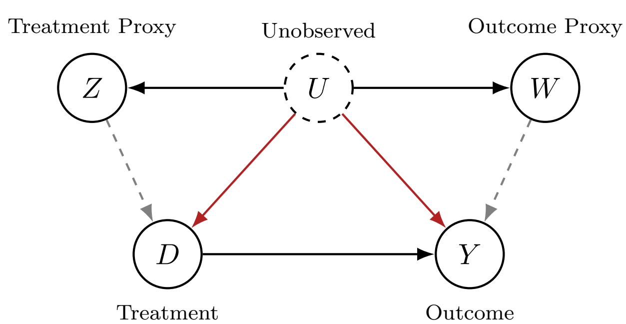
1. A **treatment proxy** Z (a.k.a. **negative control treatment**): a variable that is influenced by the unobserved confounder U and is (potentially) related to the treatment D, but is un-correlated to the outcome, once we condition on the treatment D and the confounder U.

2. An **outcome proxy** W (a.k.a. **negative control outcome**): a variable that is influenced by the unobserved confounder U and is (potentially) related to the outcome Y, but is un-affected by the treatment once we condition on the unobserved confounder.

The treatment proxy Z is an incomplete mediator on the path from U to D, while the outcome proxy W is an incomplete mediator on the path from U to Y. Neither proxy alone is sufficient to block the confounding path, but together, we will see that they can provide enough information to identify the causal effect.

4.1 The Proximal Causal Inference DAG

The causal structure underlying proximal inference is captured by the following representative directed acyclic graph:



The dashed gray arrows from Z to D and from W to Y indicate that these paths may or may not exist—the framework does not require them to be absent or present. The red arrows highlight the direct confounding paths from U to D and from U to Y that make the proxies incomplete.

4.2 The Three Identifying Assumptions

This graph satisfies the following key conditional independence properties. Any causal graph that satisfies the following three conditional independence assumptions will support the proximal identification strategy. The one we drew above is just a representative one that one should have in mind.

Assumptions for Proximal Causal Inference

1. $Y \perp\!\!\!\perp Z \mid D, U$ *(The treatment proxy Z has no direct effect on Y)*

2. $W \perp\!\!\!\perp D \mid U$ *(The treatment D has no direct causal effect on W)*

3. $Y(d) \perp\!\!\!\perp D \mid U$ *(Conditional exogeneity would hold if U were observed)*

The first assumption states that once we know the treatment D and the true confounder U, the treatment proxy Z provides no additional information about the outcome Y. In other words, Z does not have a direct causal effect on Y. The second assumption states that the treatment D does not causally affect the outcome proxy W, once we condition on the confounder U. The third assumption is the standard conditional exogeneity condition, but stated with respect to the unobserved U: if we *could* observe U, we would be able to identify the causal effect by conditioning.

It is important to emphasize that these assumptions are not limited to the specific DAG shown above. Any causal graph from which these three conditional independencies can be derived—using, for instance, d-separation—will support proximal identification. The method we will outline next solely requires these assumptions, not a particular graphical structure.

When observed covariates X are also present and they can potentially affect all the other varaibles, the assumptions are extended to condition on X as well:

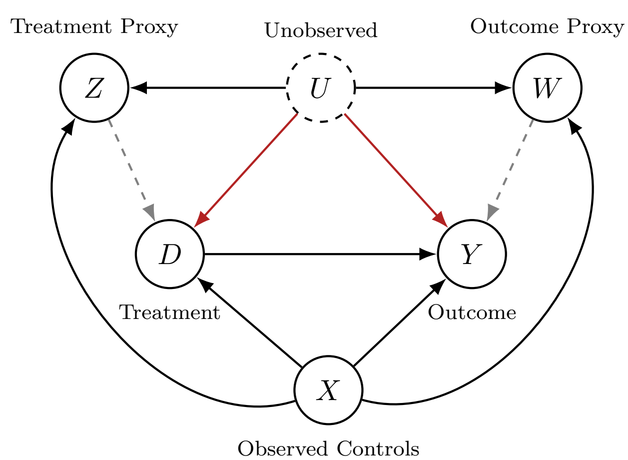
Assumptions for Proximal Causal Inference with Observed Controls X

1. $Y \perp\!\!\!\perp Z \mid D, U, X$ *(The treatment proxy Z has no direct effect on Y)*

2. $W \perp\!\!\!\perp D \mid U, X$ *(The treatment D has no direct causal effect on W)*

3. $Y(d) \perp\!\!\!\perp D \mid U, X$ *(Conditional exogeneity would hold if U were observed)*

The corresponding representative DAG with covariates includes X as an observed node with arrows to D, Y, Z, and W:



4.3 Motivating Example: HIV Drug Effectiveness

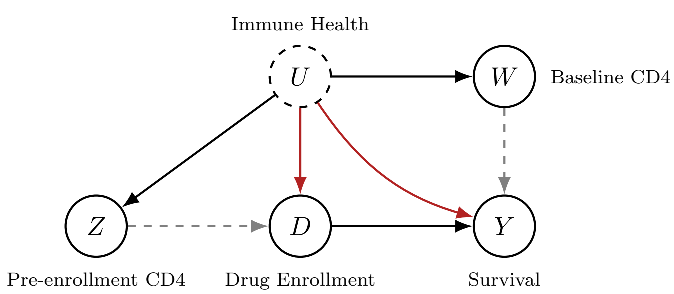
To make these abstract conditions concrete, consider the problem of evaluating the effectiveness of an HIV drug. In this setting:

- **Treatment** (D): Whether a patient is enrolled in the drug therapy.
- **Outcome** (Y): The patient's survival.
- **Unobserved confounder** (U): The patient's underlying immune health, a complex factor that a physician assesses through examination and personal communication but that is not fully captured in the data.

The physician's decision to prescribe the drug is based not only on lab measurements but also on their own clinical assessment of the patient's immune health. This clinical judgment is part of the unobserved confounding U, because it is not fully recorded in the data. As a result, there is a direct path from U to D that is not mediated by any observed variable.

Two proxies are available:

- **Treatment proxy** (Z): The pre-enrollment CD4 cell count. This lab measurement is influenced by the patient's immune health U and is one of the factors the physician uses in the treatment decision. However, it is an *incomplete* proxy because the physician also incorporates clinical judgment that goes beyond the lab measurement.
- **Outcome proxy** (W): A baseline CD4 measurement taken immediately after enrollment, before the drug has had time to take effect. This measurement reflects the patient's immune health U but is not causally affected by the treatment decision D, since it is measured at the very start of therapy.



This example satisfies the three identifying assumptions. First, once we condition on the treatment decision D and the patient's true immune health U, the pre-enrollment CD4 count Z should not provide additional information about survival Y—the lab measurement itself does not cause survival; it is the underlying immune health that matters. Second, the treatment decision D does not causally affect the baseline CD4 measurement W, because W is measured at the very beginning of therapy before the drug can take effect. Third, if we could observe the patient's true immune health U, we would be able to identify the causal effect of the drug by standard methods.

Can Z and W Be the Same Variable?

A natural question is whether the treatment proxy Z and the outcome proxy W can be the same type of measurement—for instance, both CD4 counts taken at different times. This is not a problem. Both Z and W should be thought of as noisy measurements of the true underlying immune health U. The lab measurement itself, including any measurement noise, does not affect the patient's response to the drug; it is the patient's actual immune health that matters. Therefore, Z and W can be different noisy versions of the same underlying quantity U, measured at different points in time.

Can W Have a Direct Effect on Y?

Another natural question is whether the outcome proxy W can have a direct causal effect on the outcome Y. For instance, a physician might observe the baseline CD4 result W and decide to take some other action that affects the patient's survival. The proximal framework does *not* require that W has no effect on Y—this is why the arrow from W to Y is depicted as dashed gray in the DAG rather than being absent. The key assumption is only that $W \perp\!\!\!\perp D \mid U$, not that W has no effect on Y.

4.4 Why "Negative Controls"?

The terms "negative control treatment" and "negative control outcome" have a historical origin that predates the proximal causal inference framework. Traditionally, these variables were used as **diagnostic tools** to check for the presence of unobserved confounding, rather than correct for it.

A **negative control treatment** Z is a variable that, under the assumption of no unobserved confounding, should have no effect on the outcome Y once we condition on the actual treatment D and observed covariates X. It is "negative" in the sense that its effect should be zero. If a regression reveals a statistically significant association between Z and Y after controlling for D and X, this is evidence that unobserved confounding is present—the association is being driven by the shared relationship of Z and Y with the confounder U. For instance, in HIV example, suppose that the decision to enroll was solely driven based on the pre-enrollment CD4 cell measurement, and not by any other physician consideration. Then we could check whether the prior CD4 cell measurement is un-correlated with Y conditional on treatment. If that is not the case, then we can conclude that the assumption that the treatment is solely driven by Z was wrong and there is unobserved confounding.

Similarly, a **negative control outcome** W is an outcome that the treatment D should not plausibly affect, again assuming no unobserved confounding after conditioning on covariates. In the HIV example, the drug administered today should not affect a baseline CD4 measurement taken before the treatment began. If the treatment appears to affect such a baseline outcome, it suggests the presence of unobserved confounding.

The breakthrough of the proximal causal inference literature is the realization that if both types of negative controls are available simultaneously, they can be used not merely to *detect* confounding but to *correct* for it, achieving point identification of the causal effect.

Why Not Simply Control for Z?

A natural instinct is to control for the treatment proxy Z directly, since it is correlated with the unobserved confounder U. However, Z is an *incomplete* proxy. Controlling for Z reduces confounding bias but does not eliminate it, because U retains a direct influence on D (or Y) that is not captured by Z. Before the development of proximal causal inference, the best one could do with an incomplete proxy was to use it in a sensitivity analysis to tighten the bounds. The proximal framework shows how to go further.

4.5 Empirical Applications

Proximal causal inference has been applied across a wide range of domains in recent years. The following table summarizes several notable applications, illustrating the versatility of the framework:

Application	**Treatment (D)**	**Treatment Proxy (Z)**	**Outcome Proxy (W)**	**Unobserved U**
Vaccine Effectiveness [6]	Flu vaccine	Prior-season vaccination	Off-season illness	Health-seeking behavior
Cardiovascular Effects [5]	Flu vaccine	Prior-year physician visit	Post-vaccine physician visit	Health status
Peer Effects [7]	Peer's GPA	Peer-of-peer's GPA	Peer's headaches	Homophily
Text as Proxy [8]	Policy intervention	Text features (pre)	Text features (post)	Latent context
Implicit Bias [9]	Sensitive attribute	Pre-treatment labs	Post-treatment labs	Patient health state

The peer effects application is particularly instructive. Estimating peer effects—for example, whether a friend's academic performance causally affects one's own—is notoriously difficult because of homophily: people tend to befriend others who are similar to them. Simply conditioning on friendship can open collider paths (as in the M-bias structure discussed in earlier chapters), making naive estimates unreliable. The proximal approach addresses this by using the behavior of a friend-of-a-friend as a treatment proxy (since it is influenced by the latent homophily factor but should not directly affect the individual's outcome) and a behavior of the friend that is not subject to peer effects (such as headaches) as an outcome proxy. Interestingly, the proximal analysis in [7] found no statistically significant peer effect on academic performance, even though a standard OLS approach that attempted to control for confounding would have measured a significant effect.

5 Identification in the Linear Model

We now turn to the formal identification argument. For clarity and practical relevance, we focus on the **linear setting**, which is also the most tractable case for real-world applications. The fully non-parametric extension will be discussed in Section 7.

5.1 The Linear Structural Equations

Consider the following system of linear structural equations associated with the proximal causal inference DAG (we will assume throughout that all random variables are mean-zero; have been de-meaned):

$$Y = \theta D + \beta W + \gamma U + \varepsilon_Y, \tag{1}$$

$$W = \delta U + \varepsilon_W, \tag{2}$$

where ε_Y and ε_W are structural noise terms. The structural equations for D, Z are omitted for brevity and are allowed to be fully non-parametric, i.e., $D = f_D(U, Z, \epsilon_D)$ and $Z = f_Z(U, \epsilon_Z)$. The parameter of interest is θ, the causal effect of the treatment D on the outcome Y. The coefficient β captures any direct effect of the outcome proxy W on Y, and γ captures the direct effect of the unobserved confounder U on Y.

The central challenge is the presence of U in the outcome equation (1). If we were to simply regress Y on D and W, the resulting coefficient for D would be biased because U is correlated with D. The estimate would be contaminated by the influence of the unobserved confounder.

5.2 The Substitution Trick

The key idea is to use the structural equation for W to eliminate U from the outcome equation. From equation (2), we can express the unobserved confounder as:

$$U = \frac{W - \varepsilon_W}{\delta}.$$

Substituting this into equation (1) yields:

$$\begin{aligned} Y &= \theta D + \beta W + \gamma \cdot \frac{W - \varepsilon_W}{\delta} + \varepsilon_Y \\ &= \theta D + \underbrace{\left(\beta + \frac{\gamma}{\delta}\right)}_{\kappa} W + \underbrace{\varepsilon_Y - \frac{\gamma}{\delta}\varepsilon_W}_{\eta}. \end{aligned}$$

This gives us a new equation:

$$Y = \theta D + \kappa W + \eta, \tag{3}$$

where $\kappa = \beta + \gamma/\delta$ is a composite coefficient and $\eta = \varepsilon_Y - (\gamma/\delta)\varepsilon_W$ is a new error term. The unobserved confounder U has been replaced by the observed outcome proxy W.

A natural question arises: can we now simply run an ordinary least squares (OLS) regression of Y on D and W to estimate the causal effect θ? The answer is **no**. The problem is that the new regressor W is correlated with the new error term η. This correlation exists because both W and η are functions of the same noise term ε_W. Specifically, $W = \delta U + \varepsilon_W$ contains ε_W, and $\eta = \varepsilon_Y - (\gamma/\delta)\varepsilon_W$ also contains ε_W. When a regressor is correlated with the error term, OLS produces biased and inconsistent estimates.

5.3 Why the Residual Is Uncorrelated with Z

Although OLS fails because W is correlated with η, we can leverage the treatment proxy Z to form valid moment conditions. The crucial observation is that the true residual $\eta = Y - \theta D - \kappa W$ is **uncorrelated with** Z. To see why, recall that $\eta = \varepsilon_Y - (\gamma/\delta)\varepsilon_W$. We need to show that both components are uncorrelated with Z.

Why ε_W is uncorrelated with Z: The term ε_W represents the measurement noise in the outcome proxy W. In the HIV example, this is the random noise in the post-enrollment CD4 measurement. The treatment proxy Z is the pre-enrollment CD4 count. It is reasonable to assume that the random noise in a lab measurement taken after enrollment has no relationship with the patient's CD4 count from before they entered the study. Formally, this follows because there is no causal path of influence from W to Z in the DAG and therefore, Z is independent of the idiosyncratic noise term ϵ_W associated with W.

What If Both Measurements Use the Same Lab?

A subtle point arises if both the pre-enrollment measurement Z and the post-enrollment measurement W are conducted at the same facility. If different patients use different lab facilities, and some labs are inherently noisier than others, this could introduce a correlation between the measurement errors ε_Z and ε_W through a shared "location effect." In such a case, one should include the lab facility as an observed covariate X and condition on it. After conditioning on the lab, the residual measurement noise should be independent across time points.

Why ε_Y is uncorrelated with Z: The term ε_Y is the structural noise in the outcome equation. Since, there is no causal path of influence from Y to Z, the idiosyncratic noise term associated with Y is independent of Z.

Since both ε_Y and ε_W are uncorrelated with Z, their linear combination η is also uncorrelated with Z. This gives us the first moment condition:

$$\mathbb{E}[(Y - \theta D - \kappa W) \cdot Z] = 0. \tag{4}$$

5.4 Why the Residual Is Uncorrelated with D

We have established that the residual η is uncorrelated with the treatment proxy Z. But what else is η uncorrelated with? Examining the causal graph, there is no direct path from the error terms ε_Y and ε_W to the treatment variable D. The noise terms are realized after the treatment decision has been made. This gives us a second moment condition:

$$\mathbb{E}[(Y - \theta D - \kappa W) \cdot D] = 0. \tag{5}$$

5.5 The Moment Equations and Identification

Combining equations (4) and (5), we obtain a system of two equations in two unknowns (θ and κ):

$$\mathbb{E}\left[\begin{pmatrix} D \\ Z \end{pmatrix} (Y - \theta D - \kappa W)\right] = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{6}$$

This can be written in matrix form as:

$$\underbrace{\mathbb{E}\left[\begin{pmatrix} D \\ Z \end{pmatrix} \begin{pmatrix} D & W \end{pmatrix}\right]}_{\text{Covariance matrix}} \begin{pmatrix} \theta \\ \kappa \end{pmatrix} = \mathbb{E}\left[\begin{pmatrix} D \\ Z \end{pmatrix} Y\right]. \tag{7}$$

As long as the covariance matrix on the left-hand side is **full rank**, this system has a unique solution. The full-rank condition is a **relevance condition**: it requires that both Z and W carry sufficient information about the unobserved confounder U. If Z had no relationship with U, or if W had no relationship with U, the matrix would be rank-deficient and the system would not be solvable. In practice, if the minimum eigenvalue of this matrix is very small, it indicates that the proxies are only weakly related to the confounder, leading to imprecise estimates—analogous to the weak instrument problem in IV.

The Coefficient κ Has No Causal Interpretation

While the system identifies both θ and κ, only θ has a causal interpretation as the effect of D on Y. The coefficient κ is a composite of the direct effect of W on Y (the parameter β) and the confounding effect that arises from the relationship between W and Y through U (the term γ/δ). Even though κ is "corrupted" by confounding, the coefficient θ in front of D is uncontaminated and can be trusted as the causal estimate.

5.6 Connection to Instrumental Variables

The moment equations (7) reveal a deep connection between proximal causal inference and instrumental variables. In the IV framework, we seek a linear model of the outcome whose residuals are uncorrelated with the *instruments*. In the proximal framework, we are doing exactly the same thing, but with a specific choice of "treatments" and "instruments":

Proximal Inference as Generalized IV

Proximal causal inference in the linear setting is equivalent to a **multivariate instrumental variable regression** where:

- The **endogenous variables** ("treatments") are (D, W).
- The **instruments** are (D, Z).

The causal effect θ is the coefficient on D in this generalized IV regression.

This connection is powerful for several reasons. First, it means that all the estimation machinery developed for IV—including DML-based approaches for handling high-dimensional controls—can be directly applied to the proximal setting. Second, it provides intuition: just as in IV we use an exogenous source of variation to "instrument" for the endogenous treatment, in proximal inference we use the treatment proxy Z to "instrument" for the outcome proxy W, which is correlated with the error term.

A natural question is whether it is problematic for D to appear as both an endogenous variable and an instrument. The answer is no, as long as the orthogonality conditions are satisfied. This is analogous to the standard IV setting with observed controls X: in the moment condition $\mathbb{E}[(Y - \theta D - \beta' X) \cdot (Z; X)] = 0$, the controls X appear on both sides. In the proximal setting, once we include W in the model, D becomes uncorrelated with the residual η, so it can validly serve as its own instrument.

The solution to the moment system is the **multivariate IV formula**:

$$\begin{pmatrix} \hat{\theta} \\ \hat{\kappa} \end{pmatrix} = \left(\mathbb{E}\left[\begin{pmatrix} D \\ Z \end{pmatrix} \begin{pmatrix} D & W \end{pmatrix} \right] \right)^{-1} \mathbb{E}\left[\begin{pmatrix} D \\ Z \end{pmatrix} Y \right] = \mathrm{Cov}((D;Z),(D;W))^{-1}\,\mathrm{Cov}((D;Z),Y). \qquad (8)$$

This is a generalization of the simple IV ratio $\hat{\theta} = \mathrm{Cov}(Z,Y)/\mathrm{Cov}(Z,D)$. The scalar division is replaced by multiplication with the inverse of a covariance matrix, and the scalar covariance with Y is replaced by a vector of covariances.

6 Estimation via Double Machine Learning

The connection between proximal causal inference and instrumental variables is not merely conceptual—it has immediate practical consequences. It means that the Double Machine Learning (DML) framework, which we developed for IV estimation in the previous chapter, can be applied directly to the proximal setting. This section develops the DML-Proximal estimator in detail.

6.1 Adding Covariates: The Partially Linear Model

In practice, we typically observe a set of covariates X that we wish to control for. As in the IV setting, we adopt a **partially linear model**: the outcome Y is linear in the treatment D and the outcome proxy W, but the dependence on X is left unrestricted. Similalry, the outcome proxy W is partially linear in U and X.[1]

$$\begin{aligned}
Y &= \theta D + \beta W + \gamma U + g_Y(X, \epsilon_Y) + \epsilon_Y, \\
W &= \delta U + g_W(X) + \epsilon_W \\
D &= f_D(Z, X, U, \epsilon_D) \\
Z &= f_Z(X, U, \epsilon_Z)
\end{aligned}$$

where $g_Y(X)$ and $g_W(X)$ absorb all the non-linear dependence on X.

The DML strategy is to first **residualize** all variables by subtracting the part that is predictable from X. Define the residuals:

$$\begin{aligned}
\tilde{Y} &= Y - \mathbb{E}[Y \mid X], \\
\tilde{D} &= D - \mathbb{E}[D \mid X], \\
\tilde{W} &= W - \mathbb{E}[W \mid X], \\
\tilde{Z} &= Z - \mathbb{E}[Z \mid X].
\end{aligned}$$

After residualization, the structural equations are identical to the linear setting from the previous section:

$$\begin{aligned}
\tilde{Y} &= \theta \tilde{D} + \beta \tilde{W} + \gamma \tilde{U} + \epsilon_Y, \\
\tilde{W} &= \delta \tilde{U} + \epsilon_W
\end{aligned}$$

and the exact same reasoning applies. For instance, we can apply the substitution trick to get the equation:

$$\tilde{Y} = \theta \tilde{D} + \kappa \tilde{W} + \eta,$$

with η being un-correlated with $\tilde{Z}$ and $\tilde{D}$. Hence, the moment equations take the same form as in the simple case without covariates:

$$\mathbb{E}\left[\begin{pmatrix} \tilde{D} \\ \tilde{Z} \end{pmatrix} (\tilde{Y} - \theta \tilde{D} - \kappa \tilde{W})\right] = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{9}$$

Just as the DML estimator for IV is insensitive to the first-stage models used for residualization, the proximal estimator built on this principle is also insensitive to the specific models used to predict and subtract the influence of X. This robustness is a key advantage, allowing us to use flexible machine learning methods—random forests, neural networks, gradient boosted trees—for the residualization step without invalidating the final causal estimate and confidence interval.

[1] The reasoning we develop here would also hold if we allow ϵ_Y, ϵ_W to enter the non-linear function, i.e., $g_Y(X, \epsilon_Y), g_W(X, \epsilon_W)$.

6.2 The DML-Proximal Estimator

The practical implementation of the DML-Proximal estimator proceeds as follows. First, compute **cross-fitted estimated residuals** $\check{Y}, \check{D}, \check{W}, \check{Z}$ for Y, D, W, and Z. That is, split the data into K folds, and for each fold, use the remaining $K-1$ folds to train a model predicting each variable from X, then compute the residuals on the held-out fold. Crucially, all four models must use the **same fold partition** to ensure that the cross-fitting is coupled.

Second, solve the empirical version of the moment equations (9):

$$\begin{pmatrix} \hat{\theta} \\ \hat{\kappa} \end{pmatrix} = \mathbb{E}_n \left[\begin{pmatrix} \check{D}_i \\ \check{Z}_i \end{pmatrix} \begin{pmatrix} \check{D}_i & \check{W}_i \end{pmatrix} \right]^{-1} \mathbb{E}_n \left[\begin{pmatrix} \check{D}_i \\ \check{Z}_i \end{pmatrix} \check{Y}_i \right]. \tag{10}$$

Note that Z and W can be **vectors**—there is no requirement that they be scalar. When Z and W have the same dimension, the system is exactly identified (as many equations as unknowns). When $\dim(Z) > \dim(W)$, the system is overidentified, and additional steps are needed (see Section 6.5).

6.3 Influence Function and Asymptotic Variance

A key advantage of the DML framework is that the resulting estimator $\hat{\theta}$ is **asymptotically normal**, which allows for straightforward construction of confidence intervals. The asymptotic distribution is characterized by the **influence function**.

Define the "instrument" and "treatment" vectors as:

$$R_i = \begin{pmatrix} \tilde{D}_i \\ \tilde{Z}_i \end{pmatrix}, \qquad T_i = \begin{pmatrix} \tilde{D}_i \\ \tilde{W}_i \end{pmatrix}.$$

The moment condition can be written compactly as $\mathbb{E}[R_i(\tilde{Y}_i - T_i'\boldsymbol{\beta})] = 0$, where $\boldsymbol{\beta} = (\theta, \kappa)'$. The influence function for $\boldsymbol{\beta}$ is:

$$\psi_i = \left(\mathbb{E}[R_i T_i']\right)^{-1} R_i \cdot (\tilde{Y}_i - T_i'\boldsymbol{\beta}). \tag{11}$$

The asymptotic variance of $\hat{\boldsymbol{\beta}}$ is then:

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}(0, V), \qquad V = \mathbb{E}[\psi_i \psi_i']. \tag{12}$$

In practice, we estimate V using the **sandwich formula**:

$$\hat{V} = \hat{A}^{-1}\hat{\Omega}(\hat{A}^{-1})', \tag{13}$$

where:

$$\hat{A} = \frac{1}{n}\sum_{i=1}^{n} \hat{R}_i \hat{T}_i', \qquad \hat{R}_i = \begin{pmatrix} \check{D}_i \\ \check{Z}_i \end{pmatrix}, \qquad \hat{T}_i = \begin{pmatrix} \check{D}_i \\ \check{W}_i \end{pmatrix}.$$

$$\hat{\Omega} = \frac{1}{n}\sum_{i=1}^{n} \hat{R}_i \hat{R}_i' \hat{\varepsilon}_i^2,$$

and $\hat{\varepsilon}_i = \tilde{Y}_i - T_i'\hat{\boldsymbol{\beta}}$ is the estimated residual. The standard error for $\hat{\theta}$ is then $\text{se}(\hat{\theta}) = \sqrt{\hat{V}_{11}/n}$.

This variance formula is a generalization of the familiar sandwich variance from OLS. If the instruments and treatments were the same (i.e., $R_i = T_i$), it would reduce to the standard heteroskedasticity-robust variance estimator. In the proximal setting, the mismatch between R_i and T_i reflects the fact that we are using different variables as instruments than as regressors.

6.4 Python Implementation: DML-Proximal

The following pseudocode implements the DML-Proximal estimator with coupled cross-fitting and support for vector-valued Z and W. The implementation follows the same structure as the DML-IV estimator from the previous chapter, with the key difference that the endogenous variables are (D, W) and the instruments are (D, Z).

Python: DML-Proximal Estimator (Exact Identification)

```python
import numpy as np
from sklearn.model_selection import KFold
from sklearn.base import clone

def dml_proximal(Y, D, W, Z, X, mdl_y, mdl_d, mdl_w, mdl_z, n_splits=5):
    """DML-Proximal estimator with coupled cross-fitting.
    Y: outcome (n,), D: treatment (n,),
    W: outcome proxy (n, p_w), Z: treatment proxy (n, p_z),
    X: covariates (n, p_x).  Requires dim(Z) == dim(W)."""
    n = len(Y)
    # Ensure W, Z are 2D arrays
    W = np.atleast_2d(W).reshape(n, -1)
    Z = np.atleast_2d(Z).reshape(n, -1)
    p_w, p_z = W.shape[1], Z.shape[1]
    assert p_z == p_w, "Exact identification requires dim(Z)==dim(W)"

    # Initialize residual arrays
    res_y, res_d = np.zeros(n), np.zeros(n)
    res_w, res_z = np.zeros((n, p_w)), np.zeros((n, p_z))
    # Step 1: Coupled cross-fitting -- same folds for all models
    kf = KFold(n_splits=n_splits, shuffle=True, random_state=42)
    for train_idx, test_idx in kf.split(X):
        # Fit nuisance models on training fold
        m_y = clone(mdl_y).fit(X[train_idx], Y[train_idx])
        m_d = clone(mdl_d).fit(X[train_idx], D[train_idx])
        # Fit one model per column of W and Z
        m_w = [clone(mdl_w).fit(X[train_idx], W[train_idx, j]) for j in range(p_w)]
        m_z = [clone(mdl_z).fit(X[train_idx], Z[train_idx, j]) for j in range(p_z)]
        # Compute residuals on held-out test fold
        res_y[test_idx] = Y[test_idx] - m_y.predict(X[test_idx])
        res_d[test_idx] = D[test_idx] - m_d.predict(X[test_idx])
        for j in range(p_w):
            res_w[test_idx, j] = (W[test_idx, j] - m_w[j].predict(X[test_idx]))
        for j in range(p_z):
            res_z[test_idx, j] = (Z[test_idx, j] - m_z[j].predict(X[test_idx]))

    # Step 2: Form instrument and treatment matrices
    # R = (res_d, res_z): instruments, shape (n, 1+p_z)
    # T = (res_d, res_w): treatments, shape (n, 1+p_w)
    R = np.column_stack([res_d, res_z])
    T = np.column_stack([res_d, res_w])
    # Step 3: Solve moment equations
    A = R.T @ T / n          # (1+p_z) x (1+p_w)
    b = R.T @ res_y / n      # (1+p_z,)
    coef = np.linalg.solve(A, b)
    theta_hat = coef[0]
    # Step 4: Influence function & sandwich standard errors
    eps = res_y - T @ coef                  # residuals (n,)
    Ainv = np.linalg.inv(A)                 # (1+p_z) x (1+p_w)
    psi = (Ainv @ R.T) * eps[np.newaxis, :]  # (1+p_w) x n
    V = (psi @ psi.T) / n                   # sandwich variance
    se_theta = np.sqrt(V[0, 0] / n)
    return theta_hat, se_theta
```

Several features of this implementation deserve emphasis. First, the `KFold` object is created once and reused for all four residualization models (Y, D, W, Z), ensuring that the cross-fitting is **coupled**—all models use the same fold partition. Second, the code handles **vector-valued** W and Z by fitting a separate model for each column and stacking the residuals. Third, the standard error is computed using the sandwich formula derived from the influence function, providing valid inference even under heteroskedasticity.

6.5 Handling More Instruments than Proxies

In many applications of proximal inference, the number of treatment proxies (the dimension of Z) exceeds the number of outcome proxies (the dimension of W). For instance, in a medical setting, one might have access to many pre-treatment lab measurements as treatment proxies but only a few post-treatment measurements as outcome proxies. This creates an **overidentified** system: there are more moment equations than unknowns.

At the population level, the system of equations still has a solution. However, when we move to the empirical setting and try to solve the sample analog, it is very likely that no exact solution exists. The standard approach is to create new instruments as linear combinations of the original instruments. The linear combination is chosen by "**projecting**" the outcome proxies onto the instruments, reducing the effective number of instruments to match the number of outcome proxies.

Specifically, we compute the best linear predictor of $\tilde{W}$ using $\tilde{Z}$:

$$B = \arg\min_B \mathbb{E}[\|\tilde{W} - B'\tilde{Z}\|_2^2] \Rightarrow B = \left(\mathbb{E}[\tilde{Z}\tilde{Z}']\right)^{-1} \mathbb{E}[\tilde{Z}\tilde{W}'],$$

and define the projected instrument $\bar{Z} = B'\tilde{Z}$. This new instrument has the same dimension as W and represents the best linear combination of the original instruments for predicting W. Crucially, $\bar{Z}$ is a function of the original instruments Z, so if the model's residuals are uncorrelated with Z, they are also uncorrelated with $\bar{Z}$.

If we have a large number of treatment proxies (instruments) Z, then it is important to compute the projection matrix B in a **cross-fitted** manner too, to avoid overfitting bias. Learning B on the same data used for the moment equations could introduce overfitting: the projected instruments would carry information about the realized samples W_i that we do not want. The recommended approach is to compute B out-of-fold and then use it to project $\tilde{Z}$ in the held-out fold. If, however, the number of instruments Z is small, then the matrix B can be computed using all of the data, after the residuals have been computed in a cross-fitting manner.

Python: DML-Proximal with Overidentification with Low-Dim Z

```python
def dml_proximal_overid(Y, D, W, Z, X, mdl_y, mdl_d, mdl_w, mdl_z, n_splits=5):
    """DML-Proximal with dim(Z) >= dim(W).
    Projects W onto Z via cross-fitted OLS to handle
    overidentification."""
    n = len(Y)
    W = np.atleast_2d(W).reshape(n, -1)
    Z = np.atleast_2d(Z).reshape(n, -1)
    p_w, p_z = W.shape[1], Z.shape[1]

    # Initialize residual arrays
    res_y, res_d = np.zeros(n), np.zeros(n)
    res_w, res_z = np.zeros((n, p_w)), np.zeros((n, p_z))
    z_bar = np.zeros((n, p_w))  # projected instruments

    # Step 1: Coupled cross-fitting for residualization
    kf = KFold(n_splits=n_splits, shuffle=True, random_state=42)
    folds = list(kf.split(X))
    for train_idx, test_idx in folds:
        m_y = clone(mdl_y).fit(X[train_idx], Y[train_idx])
        m_d = clone(mdl_d).fit(X[train_idx], D[train_idx])
        m_w = [clone(mdl_w).fit(X[train_idx], W[train_idx, j]) for j in range(p_w)]
        m_z = [clone(mdl_z).fit(X[train_idx], Z[train_idx, j]) for j in range(p_z)]
        res_y[test_idx] = Y[test_idx] - m_y.predict(X[test_idx])
        res_d[test_idx] = D[test_idx] - m_d.predict(X[test_idx])
        for j in range(p_w):
            res_w[test_idx, j] = (W[test_idx, j] - m_w[j].predict(X[test_idx]))
        for j in range(p_z):
            res_z[test_idx, j] = (Z[test_idx, j] - m_z[j].predict(X[test_idx]))

    # Step 2: Projection of W onto Z
    # OLS: B = (Z'Z)^{-1} Z'W, shape (p_z, p_w)
    B = np.linalg.lstsq(res_z, res_w, rcond=None)[0]
    # Project Z residuals
    z_bar = res_z @ B

    # Step 3: Solve moment equations using z_bar
    R = np.column_stack([res_d, z_bar])   # (n, 1+p_w)
    T = np.column_stack([res_d, res_w])   # (n, 1+p_w)
    A = R.T @ T / n
    b = R.T @ res_y / n
    coef = np.linalg.solve(A, b)
    theta_hat = coef[0]

    # Step 4: Influence function & sandwich standard errors
    eps = res_y - T @ coef
    Ainv = np.linalg.inv(A)
    psi = (Ainv @ R.T) * eps[np.newaxis, :]
    V = (psi @ psi.T) / n
    se_theta = np.sqrt(V[0, 0] / n)

    return theta_hat, se_theta
```

Python: DML-Proximal with Overidentification with Moderate-Dim Z

```
def dml_proximal_overid(Y, D, W, Z, X, mdl_y, mdl_d, mdl_w, mdl_z, n_splits=5):
    """DML-Proximal with dim(Z) >= dim(W).
    Projects W onto Z via cross-fitted OLS to handle
    overidentification."""
    n = len(Y)
    W = np.atleast_2d(W).reshape(n, -1)
    Z = np.atleast_2d(Z).reshape(n, -1)
    p_w, p_z = W.shape[1], Z.shape[1]

    # Initialize residual arrays
    res_y, res_d = np.zeros(n), np.zeros(n)
    res_w, res_z = np.zeros((n, p_w)), np.zeros((n, p_z))
    z_bar = np.zeros((n, p_w))  # projected instruments

    # Step 1: Coupled cross-fitting for residualization
    kf = KFold(n_splits=n_splits, shuffle=True, random_state=42)
    folds = list(kf.split(X))
    for train_idx, test_idx in folds:
        m_y = clone(mdl_y).fit(X[train_idx], Y[train_idx])
        m_d = clone(mdl_d).fit(X[train_idx], D[train_idx])
        m_w = [clone(mdl_w).fit(X[train_idx], W[train_idx, j]) for j in range(p_w)]
        m_z = [clone(mdl_z).fit(X[train_idx], Z[train_idx, j]) for j in range(p_z)]
        res_y[test_idx] = Y[test_idx] - m_y.predict(X[test_idx])
        res_d[test_idx] = D[test_idx] - m_d.predict(X[test_idx])
        res_w_train = np.zeros((len(train), p_w))
        for j in range(p_w):
            res_w[test_idx, j] = (W[test_idx, j] - m_w[j].predict(X[test_idx]))
            res_w_train[:, j] = (W[train_idx, j] - m_w[j].predict(X[train_idx]))
        for j in range(p_z):
            res_z[test_idx, j] = (Z[test_idx, j] - m_z[j].predict(X[test_idx]))
            res_z_train[:, j] = (Z[train_idx, j] - m_z[j].predict(X[train_idx]))

        # Step 2: Cross-fitted projection of W onto Z
        # OLS: B = (Z'Z)^{-1} Z'W, shape (p_z, p_w)
        B = np.linalg.lstsq(res_z_train, res_w_train, rcond=None)[0]
        # Project test-fold Z residuals
        z_bar[test_idx] = res_z[test_idx] @ B

    # Step 3: Solve moment equations using z_bar
    R = np.column_stack([res_d, z_bar])   # (n, 1+p_w)
    T = np.column_stack([res_d, res_w])   # (n, 1+p_w)
    A = R.T @ T / n
    b = R.T @ res_y / n
    coef = np.linalg.solve(A, b)
    theta_hat = coef[0]

    # Step 4: Influence function & sandwich standard errors
    eps = res_y - T @ coef
    Ainv = np.linalg.inv(A)
    psi = (Ainv @ R.T) * eps[np.newaxis, :]
    V = (psi @ psi.T) / n
    se_theta = np.sqrt(V[0, 0] / n)

    return theta_hat, se_theta
```

The key addition in this implementation is Step 2, where the projection matrix B is computed on the training residuals and applied to the test residuals, ensuring cross-fitted projection. This prevents the projected instruments from carrying information about W beyond what is captured by the linear relationship with Z.

In general, having more treatment proxies (instruments) and more outcome proxies provides greater **identifying power**. More pre-treatment measurements allow us to capture more of the variation in the unobserved confounder U, leading to more precise estimates. In practice, it is advisable to collect as many post-treatment measurements W as the dimension of the unobserved confounder we are trying to capture and as many pre-treatment measurements as possible.

7 Beyond Partial Linearity: The Non-Parametric Case

The partially linear identification result and the DML-Proximal estimator developed in the previous sections rely on the assumption that the outcome is partially linear in the treatment, outcome proxy and confounder and that the outcome proxy is partially linear in the confounder. This section sketches how the proximal framework extends to the fully non-parametric setting, where no functional form assumptions are imposed on the structural equations.

7.1 The Non-Parametric Moment Condition

In the linear case, we sought a linear model of Y in terms of D and W such that the residuals were uncorrelated with D and Z. In the non-parametric case, we generalize this by seeking a *flexible* predictive model $h(D, W)$ such that the residuals are not merely uncorrelated with (D, Z), but **conditionally mean-zero**, conditional on any value of D, Z (this essentially asks that the residuals are **independent** of (D, Z) not just uncorrelated):

Non-Parametric Proximal Moment Condition

Find a function $h(D, W)$ such that:

$$\mathbb{E}[Y - h(D, W) \mid D, Z] = 0.$$

This is a stronger condition than uncorrelatedness: it requires that the conditional expectation of the residual, given any value of D and Z, is exactly zero. For instance, this property implies that:

$$\forall f : \mathbb{E}[(Y - h(D, W))\, f(D, Z)] = 0$$

In the linear case, we only required this to be satisfied for the linear functions $f(D, Z) = D$ and $f(D, Z) = Z$, which led to the moment conditions we derived earlier for the linear setting. Requiring this property for all functions f, essentially implies independence of the residuals with (D, Z), while requiring this property to be satisfied only for linear functions f, implies only un-correlatedness of the residuals.

7.2 The Proximal g-Formula

In the linear model, we noted that the coefficient κ on W has no causal interpretation—only the coefficient θ on D is causally meaningful. The non-parametric analog of "reading off the coefficient on D" is to apply the **g-formula** to the learned model h:

Proximal g-Formula

The Average Treatment Effect is recovered as:

$$\text{ATE} = \mathbb{E}[h(1, W) - h(0, W)].$$

This formula asks: what is the average difference in the model's predictions when we flip the treatment from $D = 0$ to $D = 1$, averaging over the distribution of the outcome proxy W? If h were a linear function $\theta D + \kappa W$, this would simply return θ. In the non-parametric case, it extracts the causal effect of D while discarding the model's behavior with respect to W, which is not causally interpretable.

When observed covariates X are present, the moment condition and g-formula are extended accordingly:

$$\mathbb{E}[Y - h(D, W, X) \mid D, Z, X] = 0, \qquad \text{ATE} = \mathbb{E}[h(1, W, X) - h(0, W, X)].$$

7.3 Why This Is Not a Standard Regression Problem

A critical point is that finding h is **not** a standard regression problem. In a standard regression, we would minimize the mean squared prediction error $\mathbb{E}[(Y - h(D, W))^2]$ with respect to h. However, this would simply yield the conditional expectation $\mathbb{E}[Y \mid D, W]$, which is a biased estimate of the causal effect—it is the non-parametric analog of OLS, and we have already seen that OLS fails in this setting.

The fundamental issue is the **mismatch** between the variables in the model and the variables in the conditioning set of the moment condition. The function h takes (D, W) as inputs, but the moment condition requires the residuals to be conditionally mean-zero given a *different* set of variables, (D, Z). If the conditioning set were also (D, W), minimizing the squared loss would be the correct approach. Because it is (D, Z), we are in the territory of a **conditional moment restriction** problem, which is fundamentally harder than standard regression. This problem is known in the literature as the **Non-Parametric Instrumental Variable (NPIV)** problem.

7.4 Completeness

For the non-parametric identification to be valid, an additional technical condition is required: the conditional moment problem must admit a solution, and the set of solutions must be "essentially unique." This is formalized through a **completeness** condition, which requires that the set of solutions to the observable moment condition $\mathbb{E}[Y - h(D,W) \mid D, Z] = 0$ coincides with the set of solutions to the unobservable condition $\mathbb{E}[Y - h(D,W) \mid D, U] = 0$. Completeness is a regularity condition that ensures the proxies Z and W carry enough information about U to pin down the causal effect.

7.5 Methods for the NPIV Problem

Solving for h such that $\mathbb{E}[Y - h(D,W) \mid D, Z] = 0$ is known as the **Non-Parametric Instrumental Variable Regression** (NPIV) problem. This is an active area of research at the intersection of econometrics and machine learning. Two representative approaches are:

Two-Stage Approach (DeepIV) [10]. The DeepIV method is based on a modified squared loss. Instead of minimizing $\mathbb{E}[(Y - h(D,W))^2]$ directly, it first learns the conditional density of (D, W) given (D, Z), and then optimizes h by averaging its predictions with respect to this learned density:

$$\min_h \; \mathbb{E}\left[\left(Y - \hat{\mathbb{E}}[h(D,W) \mid D, Z]\right)^2\right] + \lambda \mathbb{E}[h(D,W)^2].$$

The regularization term $\lambda \mathbb{E}[h(D,W)^2]$ is important for stability [11]. DeepIV typically uses deep Gaussian mixture models for the density estimation step and neural networks for the function h.

Adversarial / Minimax Approach [12]. An alternative is to frame the problem as a game between two players. The conditional moment restriction $\mathbb{E}[Y - h(D,W) \mid D, Z] = 0$ implies, by the tower rule, that $\mathbb{E}[(Y - h(D,W)) \cdot f(D,Z)] = 0$ for *any* function f. This leads to the minimax formulation:

$$\min_h \max_f \; \mathbb{E}[(Y - h(D,W)) \cdot f(D,Z)] - \mathbb{E}[f(D,Z)^2] + \lambda \mathbb{E}[h(D,W)^2].$$

A "learner" chooses h to make the residuals small, while an "adversary" (or critic) f tries to find a function of (D, Z) that reveals violations of the moment condition. The training proceeds via gradient descent on h and gradient ascent on f, similar to the training of Generative Adversarial Networks (GANs). The extra terms can be viewed as regularization terms that stabilize the critic's and the learners optimization problems and are useful both for statistical considerations (fast rates and avoiding overfitting), as well as for optimization purposes (strong convexity to the min-max objective).

Both approaches are significantly more complex than the partially linear case and remain active areas of research. The NPIV problem is inherently harder than standard regression because it involves solving an inverse problem (the solution to a "convolution problem", i.e., find function h such that $\int h(x)q(x \mid z = \cdot)dx = r(\cdot))$. For practical applications with moderate sample sizes, the partially linear DML-Proximal estimator is generally the recommended approach.

8 Comparing Identification Strategies

Having now developed three distinct strategies for dealing with unobserved confounding—the front-door criterion (Chapter 14), instrumental variables (Chapter 15), and proximal causal inference (this chapter)—it is instructive to compare them systematically. The choice of strategy is dictated by the **causal structure** of the problem at hand: what special variables are available and what paths exist in the causal graph.

8.1 A Side-by-Side Comparison

The following table summarizes the key distinguishing features of each strategy:

	Front-Door	**IV**	**Proximal**
Special variable(s)	Mediator M	Instrument Z	Treatment proxy Z, Outcome proxy W
Exclusion	D affects Y *only* through M; no unconfounded direct effect. No unobserved confounding between D and M or M and Y.	Z affects Y *only* through D; no direct $Z \to Y$. No unobserved confounding between Z and Y.	$Z \perp\!\!\!\perp Y \mid D, U$; $W \perp\!\!\!\perp D \mid U$; $Y(d) \perp\!\!\!\perp D \mid U$.
Relevance	No constraint	Z shifts D	Z, W each carry information about U
Identifies	ATE	LATE (binary D) or partial-linear ATE	ATE (partial-linear or NPIV)

8.2 Matching Examples to Strategies

To solidify the comparison, consider three examples from this and previous chapters, each naturally suited to a different strategy.

Example 1: Ride-Sharing and Tipping (Front-Door). In the study by Bellemare, Bloem, and Wexler (2024), the treatment D is authorizing a shared ride, the outcome Y is tips, and the unobserved confounder U is the rider's frugality. The platform observes whether the ride was *actually shared* (M), which depends on the algorithm matching another rider on the same route.

The variable M is a **complete, unconfounded mediator**: D affects Y only through M, and the algorithm's matching decision does not depend on the rider's unobserved frugality. This makes the front-door criterion the appropriate strategy.

Why not IV or Proximal? There is no exogenous instrument available—no random variation that shifts the treatment independently of the confounder. And there are not two distinct types of proxies for the rider's frugality.

Example 2: Oregon Medicaid Lottery (IV). In the study by Finkelstein et al. (2012), the treatment D is Medicaid coverage, the outcome Y is ER visits, and the unobserved confounder U is socioeconomic and health status. Oregon expanded Medicaid via a lottery, providing a natural instrument Z: the lottery outcome. The lottery is randomly assigned (exogenous), it shifts the probability of obtaining Medicaid (relevant), and it has no direct effect on ER visits except through coverage (exclusion). The LATE identifies the causal effect for compliers—those whose coverage status is changed by winning the lottery.

Why not Front-Door or Proximal? There is no complete mediator between Medicaid coverage and ER visits. The lottery is exogenous, not a proxy for the confounder U—it is not influenced by the patient's health status.

Example 3: HIV Drug Effectiveness (Proximal). In the study by Tchetgen Tchetgen et al. (2020), the treatment D is drug enrollment, the outcome Y is survival, and the unobserved confounder U is the patient's immune health. The pre-enrollment CD4 count (Z) is a treatment proxy, and the baseline post-enrollment CD4 count (W) is an outcome proxy. Together, they enable proximal identification.

Why not IV? The pre-enrollment CD4 count Z is *not* a valid instrument. It does not satisfy the exclusion restriction because the physician's clinical assessment of U creates another path from U to D that does not go through Z. In the IV framework, the instrument must not be confounded, but here the physician's judgment provides an additional, unobserved source of confounding between Z and Y. Why not Front-Door? There is no complete mediator $D \to M \to Y$ in this setting.

The Choice Is Dictated by the Causal Structure

There is no universal "best" strategy for dealing with unobserved confounding. Each strategy requires a different structural feature of the causal graph:

- **Front-Door** requires a complete, unconfounded mediator.
- **IV** requires a conditionally exogenous instrument that shifts the treatment.
- **Proximal** requires two types of incomplete proxies for the confounder.

The analyst must carefully examine the causal structure of their specific problem to determine which strategy, if any, is applicable.

9 Summary and Takeaways

This chapter introduced proximal causal inference, a powerful identification strategy for settings with unobserved confounding. The key ideas are:

1. **Negative controls detect AND correct.** Treatment proxies Z (negative control treatments) and outcome proxies W (negative control outcomes) were traditionally used to check for confounding. The proximal causal inference framework shows that having both types simultaneously enables **point identification** of the causal effect, going beyond the bounds provided by sensitivity analysis.

2. **Proximal inference generalizes IV.** In the partially linear setting, proximal causal inference reduces to an IV-like method where (D, W) are the "endogenous variables" and (D, Z) are the "instruments." The DML cross-fitting machinery applies directly, providing an asymptotically normal estimator with influence-function-based standard errors.

3. **Practical and widely applied.** Proximal inference has been deployed in vaccine effectiveness studies, cardiovascular health research, peer effects estimation, clinical decision-making bias detection, and settings where text data serves as proxies.

4. **Beyond linearity, NPIV methods apply.** In the fully non-parametric setting, proximal inference requires solving a Non-Parametric IV (NPIV) problem, for which modern machine learning methods (DeepIV, adversarial/minimax approaches) provide practical, if still research-level, solutions.

References

[1] Miao, W., Geng, Z., & Tchetgen Tchetgen, E. J. (2018). Identifying causal effects with proxy variables of an unmeasured confounder. *Biometrika*, 105(4), 987–993.

[2] Tchetgen, E. J. T., Ying, A., Cui, Y., Shi, X., & Miao, W. (2020). An introduction to proximal causal learning. *arXiv preprint* arXiv:2009.10982.

[3] Cui, Y., Pu, H., Shi, X., Miao, W., & Tchetgen Tchetgen, E. (2024). Semiparametric proximal causal inference. *Journal of the American Statistical Association*, 119(546), 1348-1359.

[4] Shi, X., Miao, W., Nelson, J. C., & Tchetgen Tchetgen, E. J. (2020). Multiply robust causal inference with double-negative control adjustment for categorical unmeasured confounding. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 82(2), 521-540.

[5] Guo, J., Wang, T., Liu, Z., Zeng, W., Shen, P., Sun, Y., ... & Xu, Y. (2025). Estimating cardiovascular effects of influenza vaccination in older adults: a target trial emulation using proximal causal inference. *Eclinicalmedicine*, 87.

[6] Li, K. Q., Shi, X., Miao, W., & Tchetgen Tchetgen, E. (2024). Double negative control inference in test-negative design studies of vaccine effectiveness. *Journal of the American Statistical Association*, 119(547), 1859-1870.

[7] Egami, N., & Tchetgen Tchetgen, E. J. (2024). Identification and estimation of causal peer effects using double negative controls for unmeasured network confounding. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 86(2), 487-511.

[8] Chen, J., Bhattacharya, R., & Keith, K. (2024). Proximal causal inference with text data. *Advances in Neural Information Processing Systems*, 37, 135983-136017.

[9] Liu, K., Altman, R., & Syrgkanis, V. (2024). Detecting clinician implicit biases in diagnoses using proximal causal inference. In *Biocomputing 2025: Proceedings of the Pacific Symposium* (pp. 330-345).

[10] Hartford, J., Lewis, G., Leyton-Brown, K., & Taddy, M. (2017, July). Deep IV: A flexible approach for counterfactual prediction. In *International conference on machine learning* (pp. 1414-1423). PMLR.

[11] Li, Z., Lan, H., Syrgkanis, V., Wang, M., & Uehara, M. (2024). Regularized deepiv with model selection. *arXiv preprint* arXiv:2403.04236.

[12] Dikkala, N., Lewis, G., Mackey, L., & Syrgkanis, V. (2020). Minimax estimation of conditional moment models. *Advances in Neural Information Processing Systems*, 33, 12248-12262.

[13] Chernozhukov, V., Hansen, C., Kallus, N., Spindler, M., & Syrgkanis, V. (2026). *Applied Causal Inference Powered by ML and AI.*